THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 9 & 10: **Light – Reflection and Refraction, The Human Eye and the Colourful World** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A spherical mirror and a thin spherical lens each have focal length −15 cm. They are: 1

(a) both concave mirror and concave lens **(b)** both convex
(c) concave mirror, convex lens **(d)** convex mirror, concave lens

2. The speed of light in vacuum is: 1

(a) 3×10^5 m/s **(b)** 3×10^{10} m/s **(c)** 3×10^8 m/s **(d)** 3×10^6 m/s

3. A ray parallel to the principal axis after refraction through a convex lens: 1

(a) passes through F **(b)** becomes parallel
(c) goes undeviated **(d)** passes through 2F

4. Magnification of a lens is defined as: 1

(a) u/v **(b)** $-v/u$ **(c)** f/u **(d)** $h'/h = v/u$

5. The focal length of a convex lens is 25 cm. Its power is: 1

(a) +4 D **(b)** +0.25 D **(c)** −4 D **(d)** +25 D

6. A prism has how many rectangular lateral surfaces? 1

(a) two **(b)** five **(c)** four **(d)** three

7. The defect of vision in which the near point is beyond 25 cm is: 1

(a) astigmatism **(b)** colour blindness **(c)** hypermetropia **(d)** myopia

8. Signal lights for danger are red because red is: 1

(a) brightest **(b)** most absorbed **(c)** least scattered **(d)** most scattered

9. The apparent flattening of the Sun's disc at sunrise is due to: 1

(a) reflection **(b)** scattering
(c) dispersion **(d)** atmospheric refraction

10. The eye defect presbyopia mainly occurs: 1

(a) at birth **(b)** due to bright light
(c) in children **(d)** with ageing

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is meant by lateral displacement? 2

12. Light travels from air to glass ($n = 1.5$). If the speed in air is 3×10^8 m/s, find the speed in glass. 2

13. Why does a prism split white light while a glass slab does not? 2

14. What is the colour of the sky at high altitude? Why? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why does a concave mirror work as a shaving mirror? Explain with the position of the object. 3

16. Describe how a glass prism produces a spectrum. 3

17. Find the focal length of a concave mirror if an object placed at 20 cm forms an image at 20 cm in front of it. 3

OR

Two lenses have powers +1.5 D and −0.5 D. Find the net power and the focal length of the combination. 3

18. The near point of a hypermetropic eye is 1 m. Find the power of the lens needed to correct it (normal near point 25 cm). 3

OR

What would happen if the human eye lens had no power of accommodation? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain the New Cartesian sign convention for lenses.
(ii) A concave lens has focal length 20 cm. At what distance should an object be placed so that it forms an image at 15 cm? 5

OR

(i) Draw (describe) the ray diagrams for a convex mirror for an object at infinity and at a finite distance.
(ii) Why is a convex mirror called a diverging mirror?
(iii) Give one use. 5

20. (i) Explain how a rainbow is formed.
(ii) Why is it seen only after rain?
(iii) Where should the Sun be for an observer to see it? 5

OR

(i) Explain the scattering of light.
(ii) How does the colour of scattered light depend on particle size?
(iii) Why is the sky blue and the sunset red? 5

END OF QUESTION PAPER — **Total: 40 Marks**

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